FALL 2026
CS 103000

Professor
Madeline Blount
she/her

Week 0

Dall-E 2: cats learning C++ in the forest on '90's technology

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Madeline Blount
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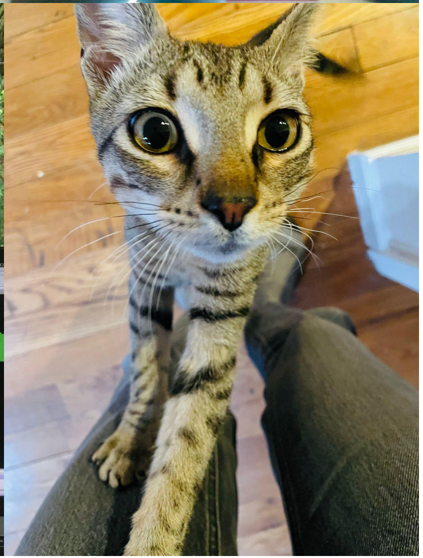
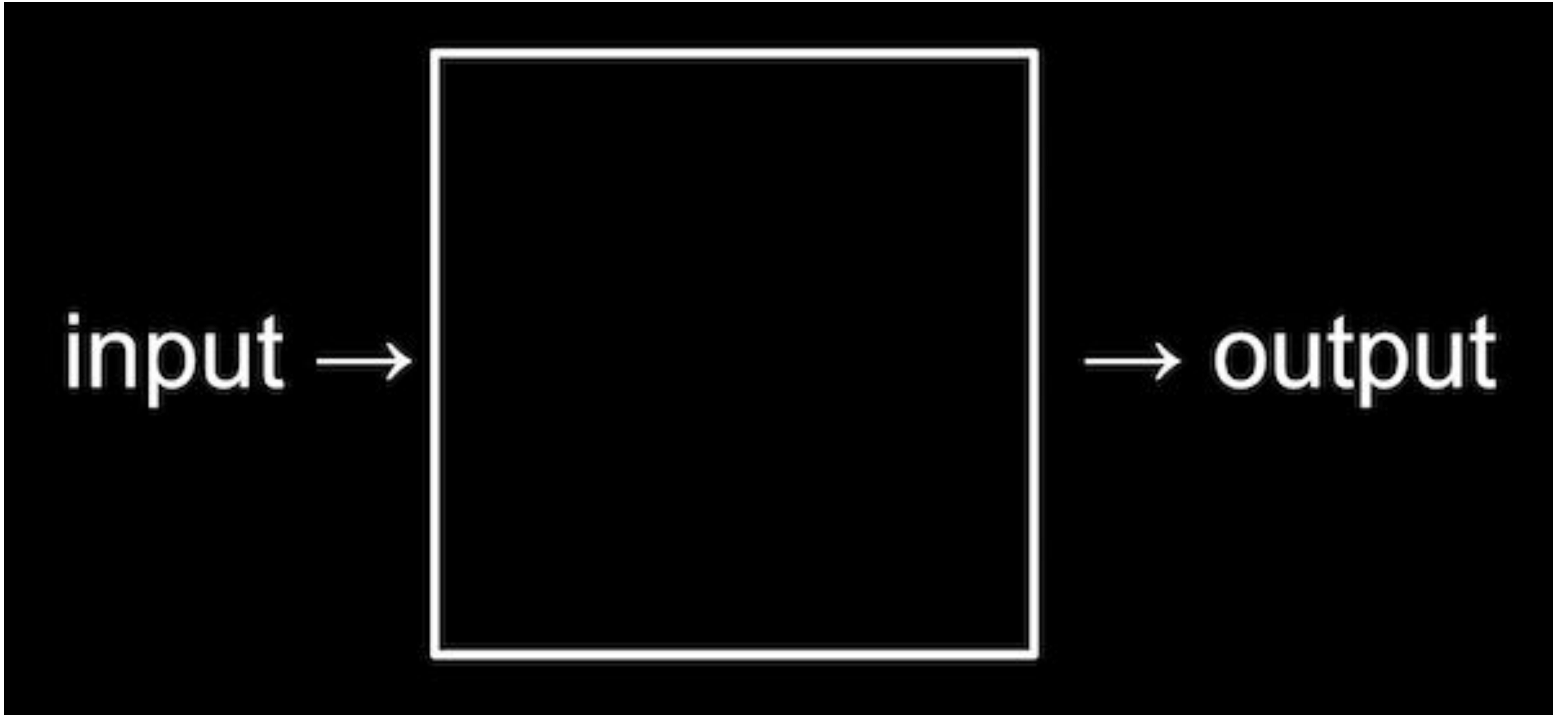


COMPUTATIONAL THINKING:



Harvard CS50

Human language:

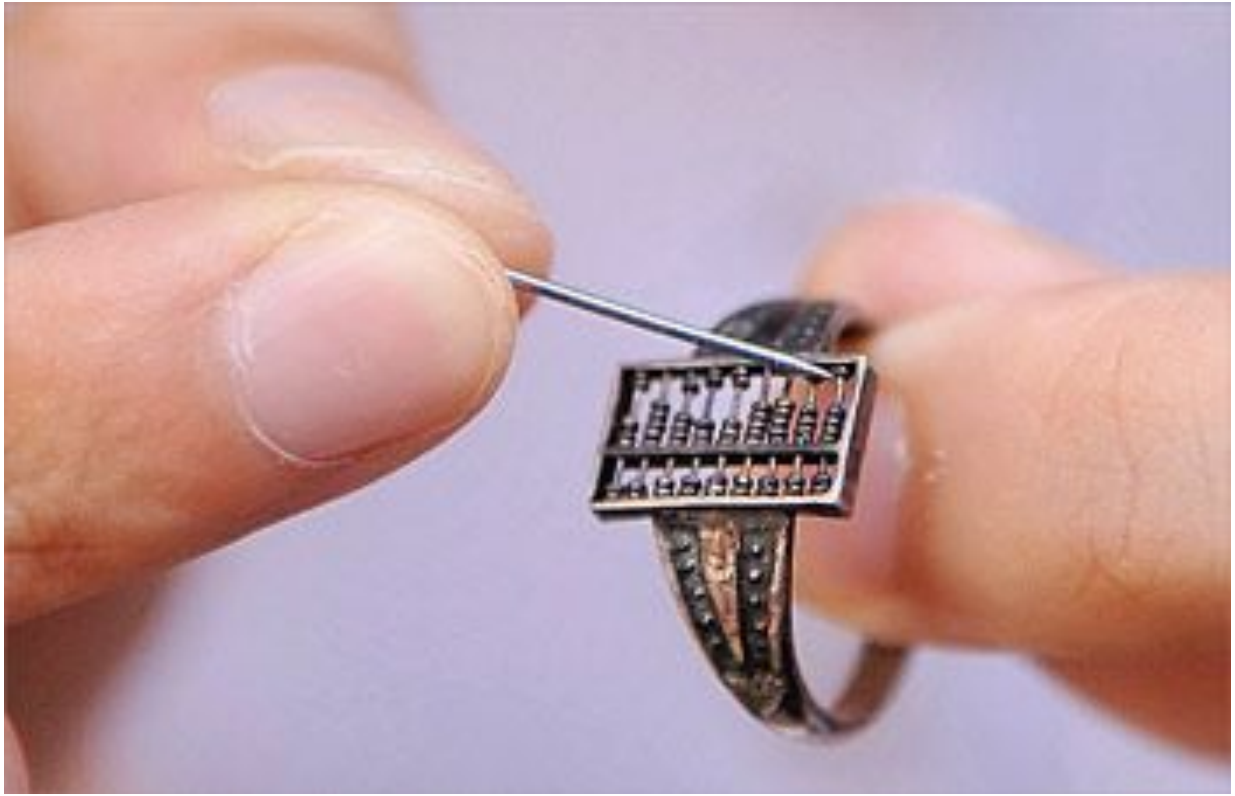
I want it to say “hello world”

High-level languages: C++

```
cout << “hello world”;
```

machine instructions: BINARY

010101110100101001001010100
101000101001010010101001010



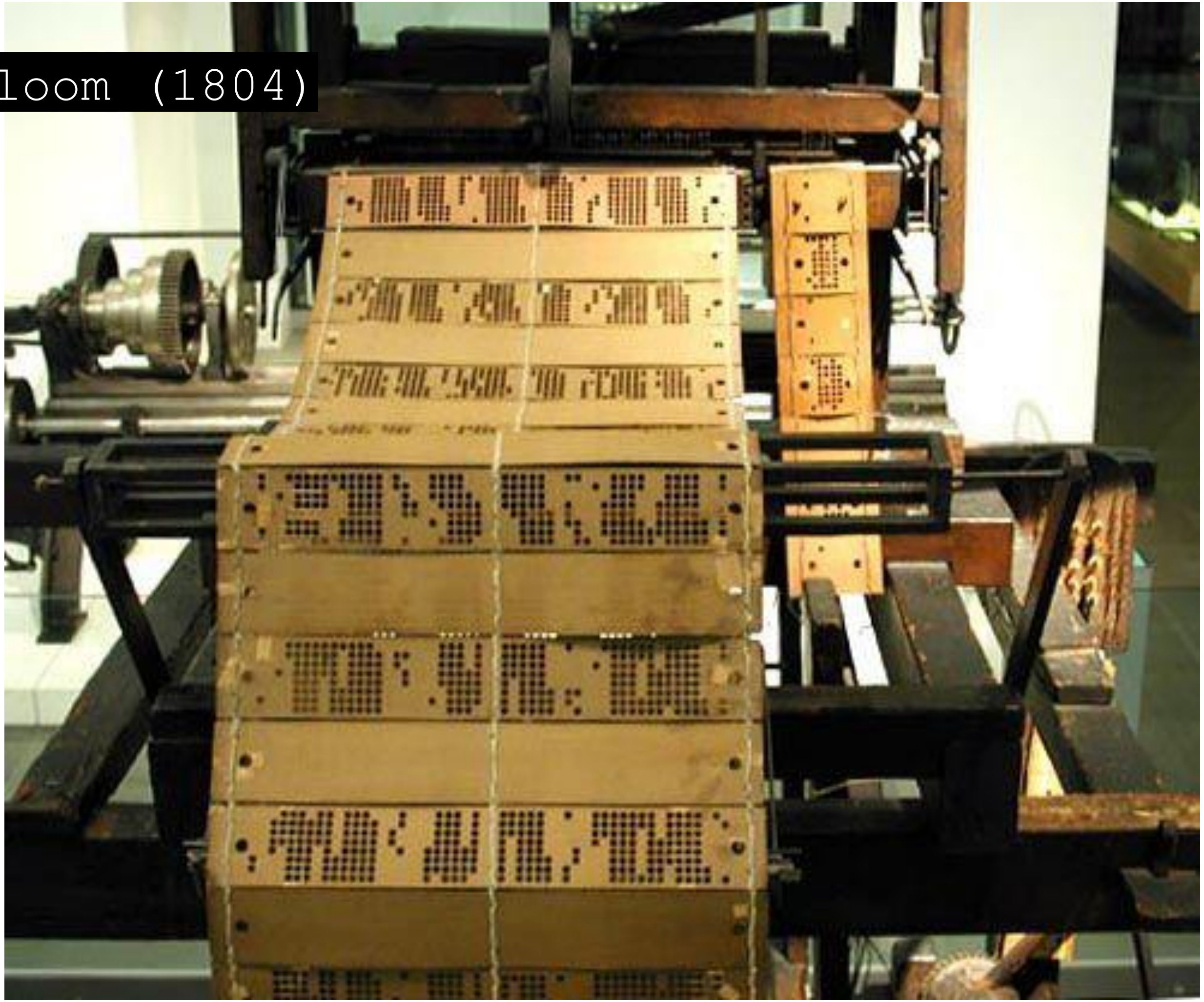
abacus "wearable": 1700

Wikimedia

textiles

Wikimedia

Jacquard loom (1804)



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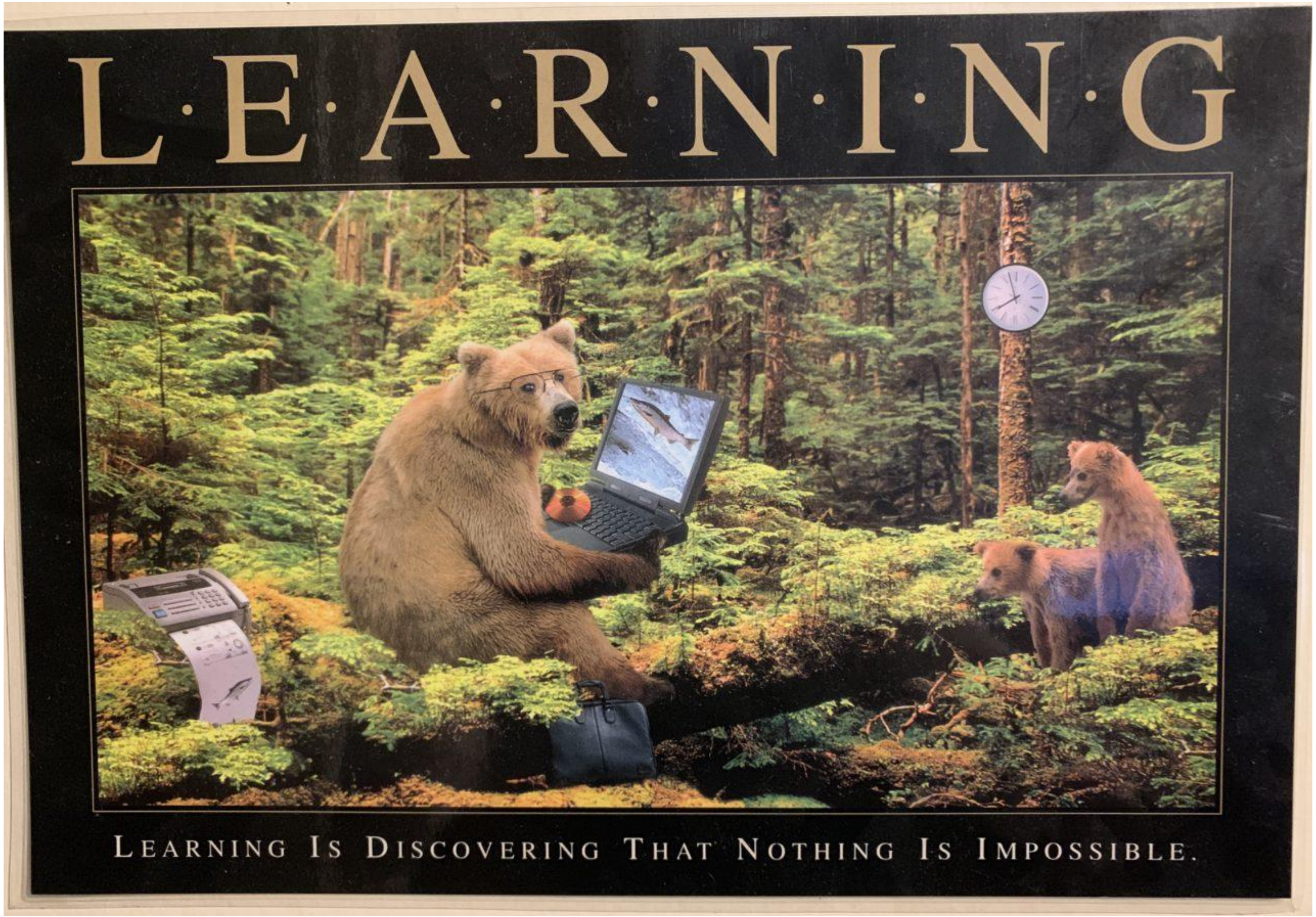

L·E·A·R·N·I·N·G
LEARNING IS DISCOVERING THAT NOTHING IS IMPOSSIBLE.

ENGLISH:

How are you?

GREEK:

τι κάνεις;

C++:

```
cout << "Hello World!" << endl;
cout << "How are you?" << endl;
```

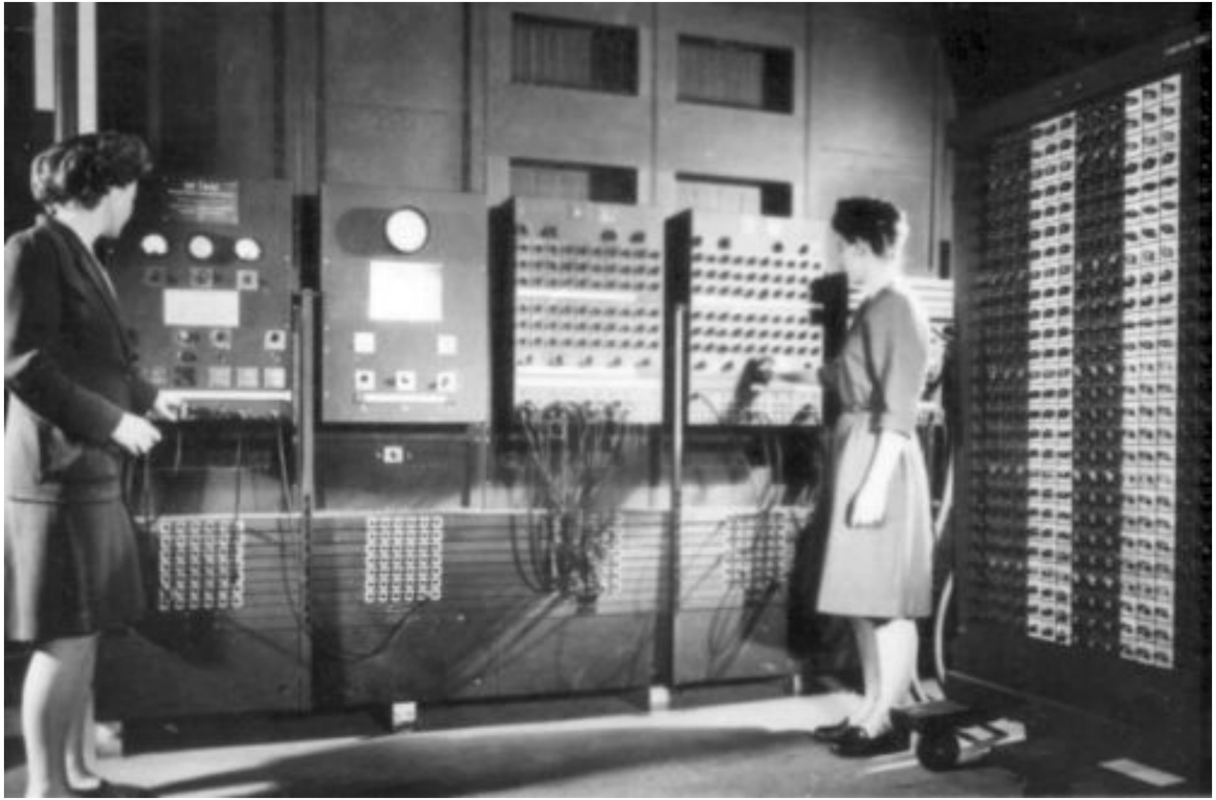
```
Hello World!
How are you?
```

BINARY (machine instructions):

01001000
00100000
01100100

(Hello World!)

early computers, 1930's-1940's



zyBooks, from Army Archives

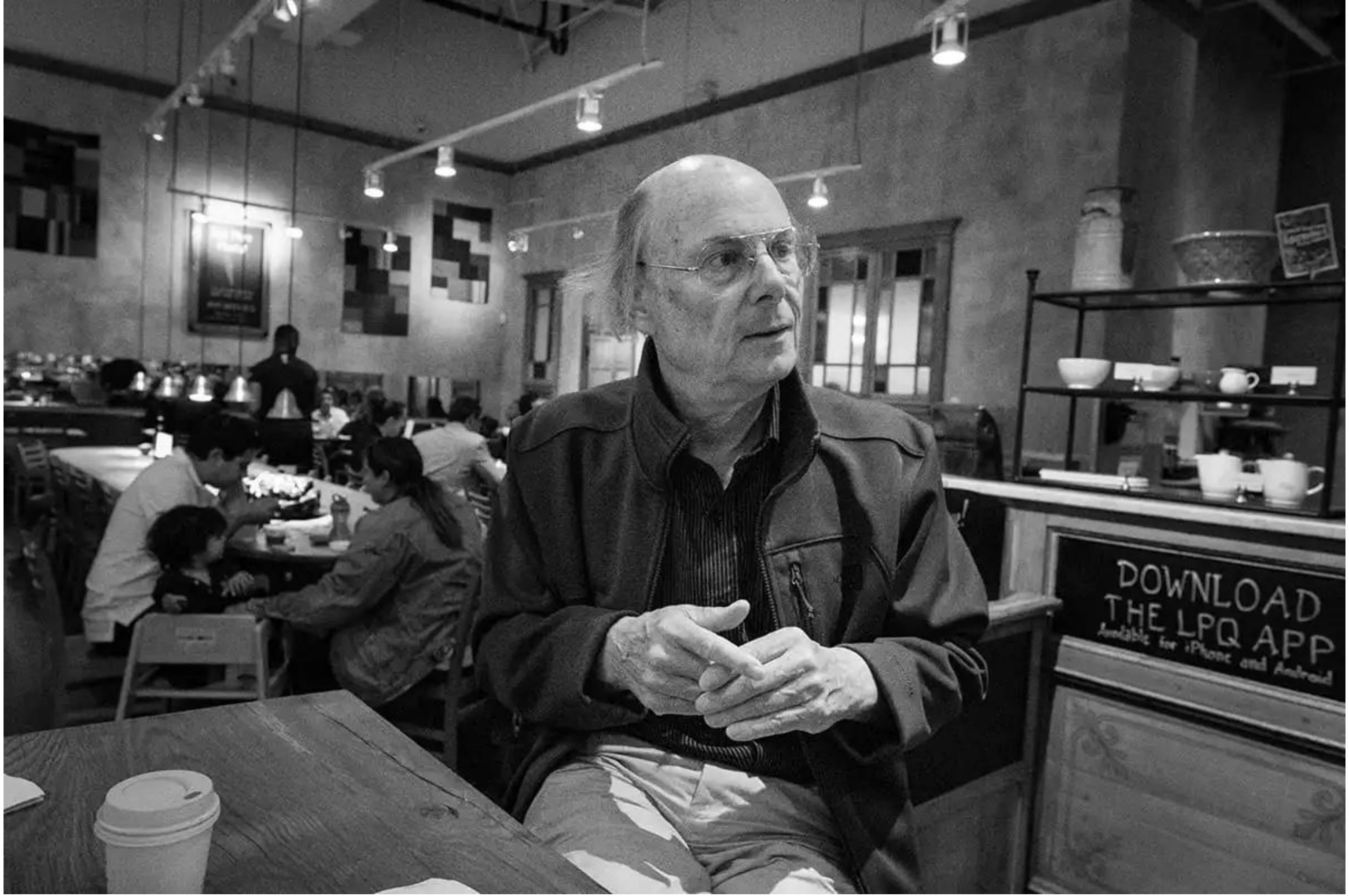
Why C++?

- Had to choose some language!!
- Good foundational language
- Fast, efficient w/memory

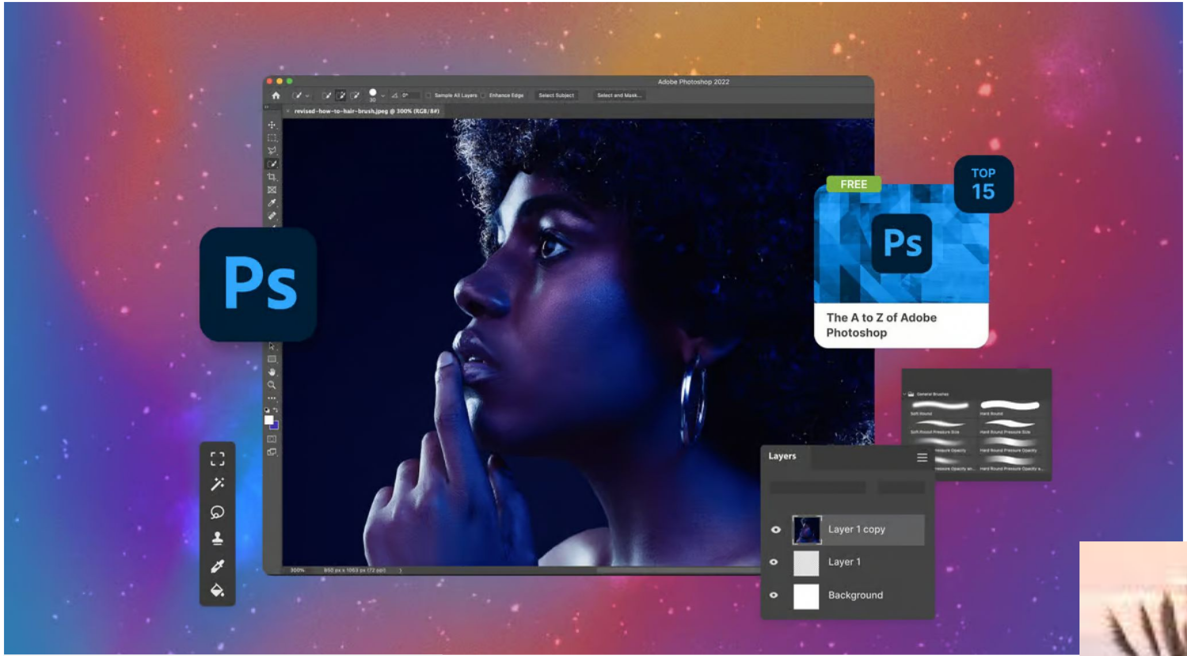
- *What we learn here can translate to other languages.*

Bjarne Stroustrup:

“The aims of C++ are noble: enable programmers to write real-world programs that are simultaneously elegant and efficient, to raise the level of abstraction in real-world code, and thereby improve the working lives of hundreds of thousands of serious programmers.”



Codecademy, 2019


Adobe Photoshop 2022
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TOP
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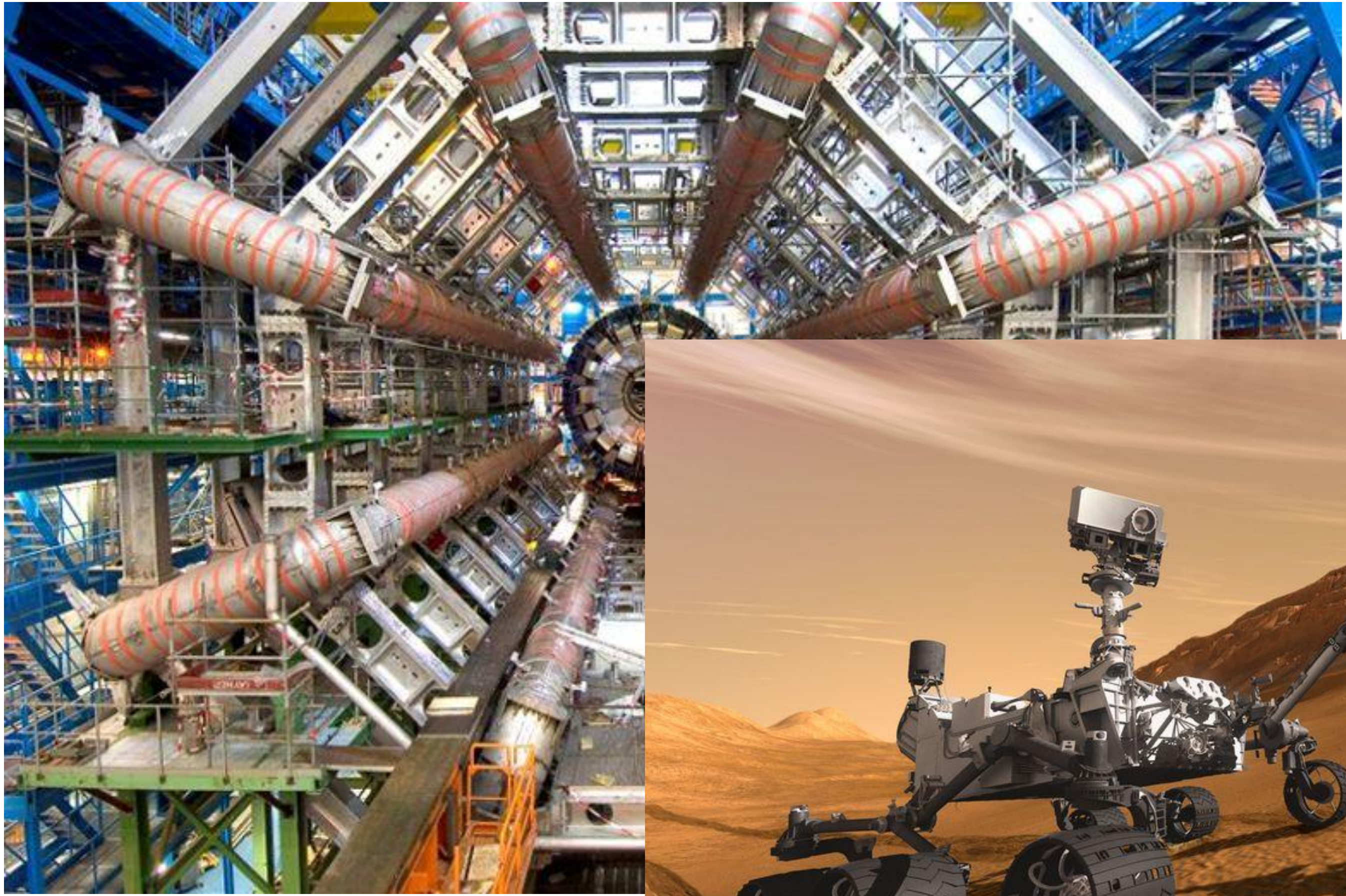
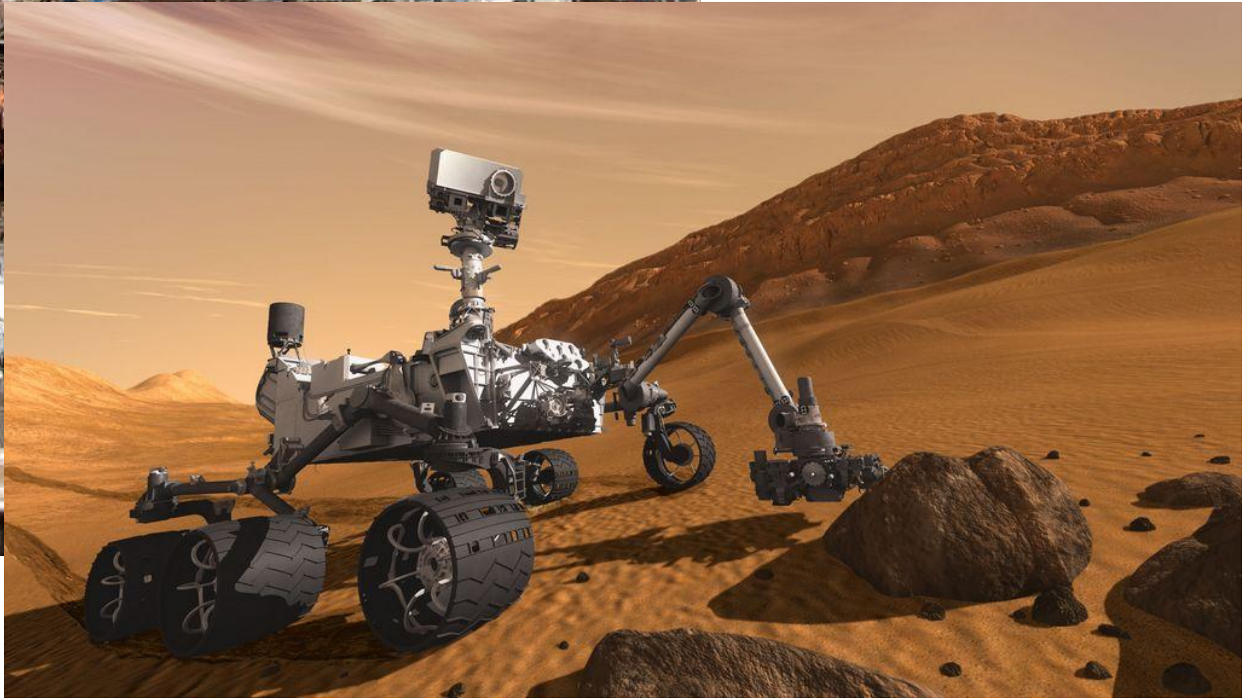
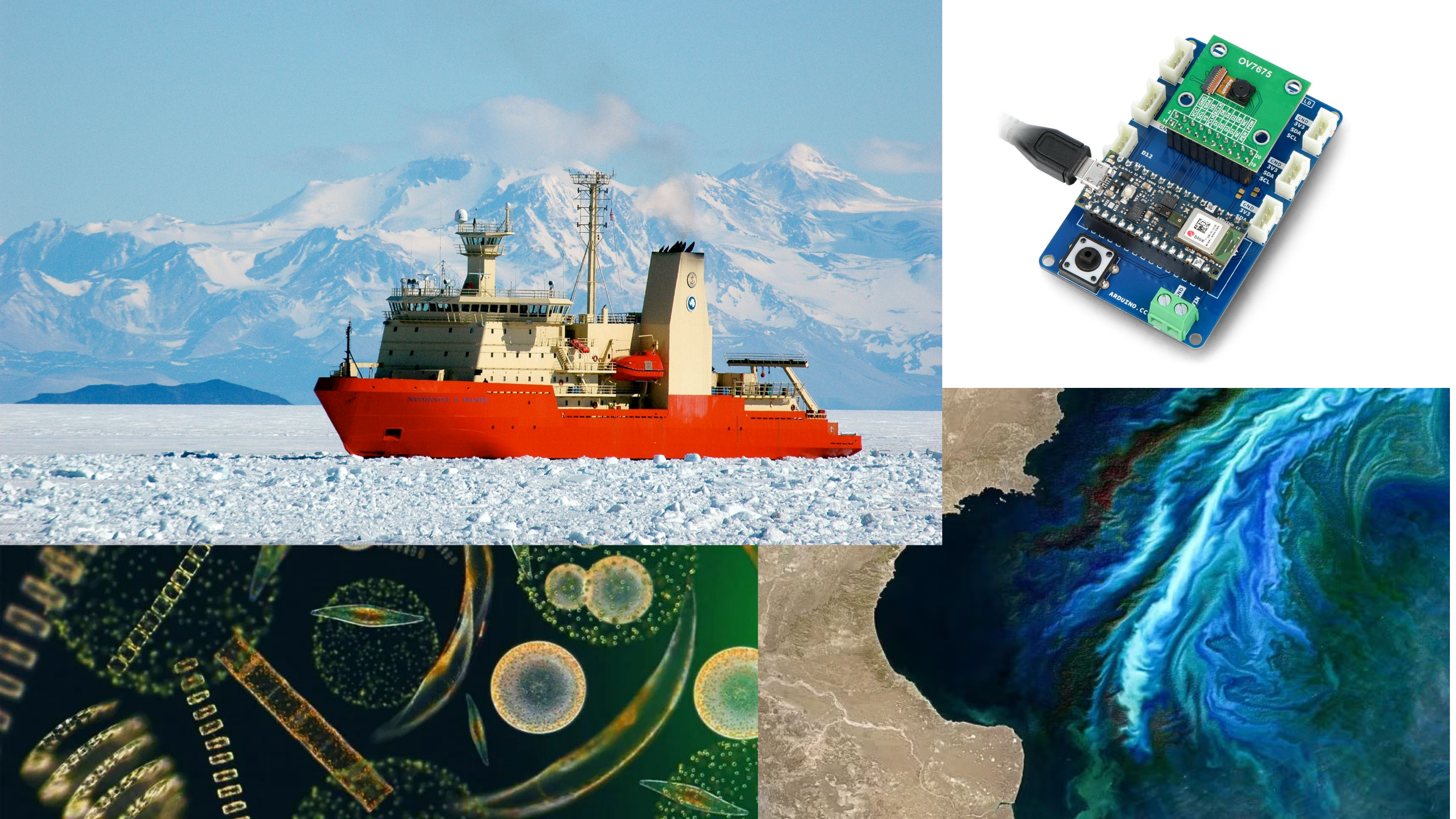
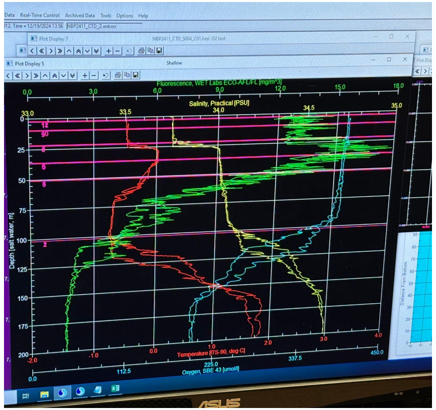
The A to Z of Adobe
Photoshop
Layers
Layer 1 copy
Layer 1
Background


unity






OV7675
ARDUINO.CC



National
Science
Foundation

Grace Hopper:

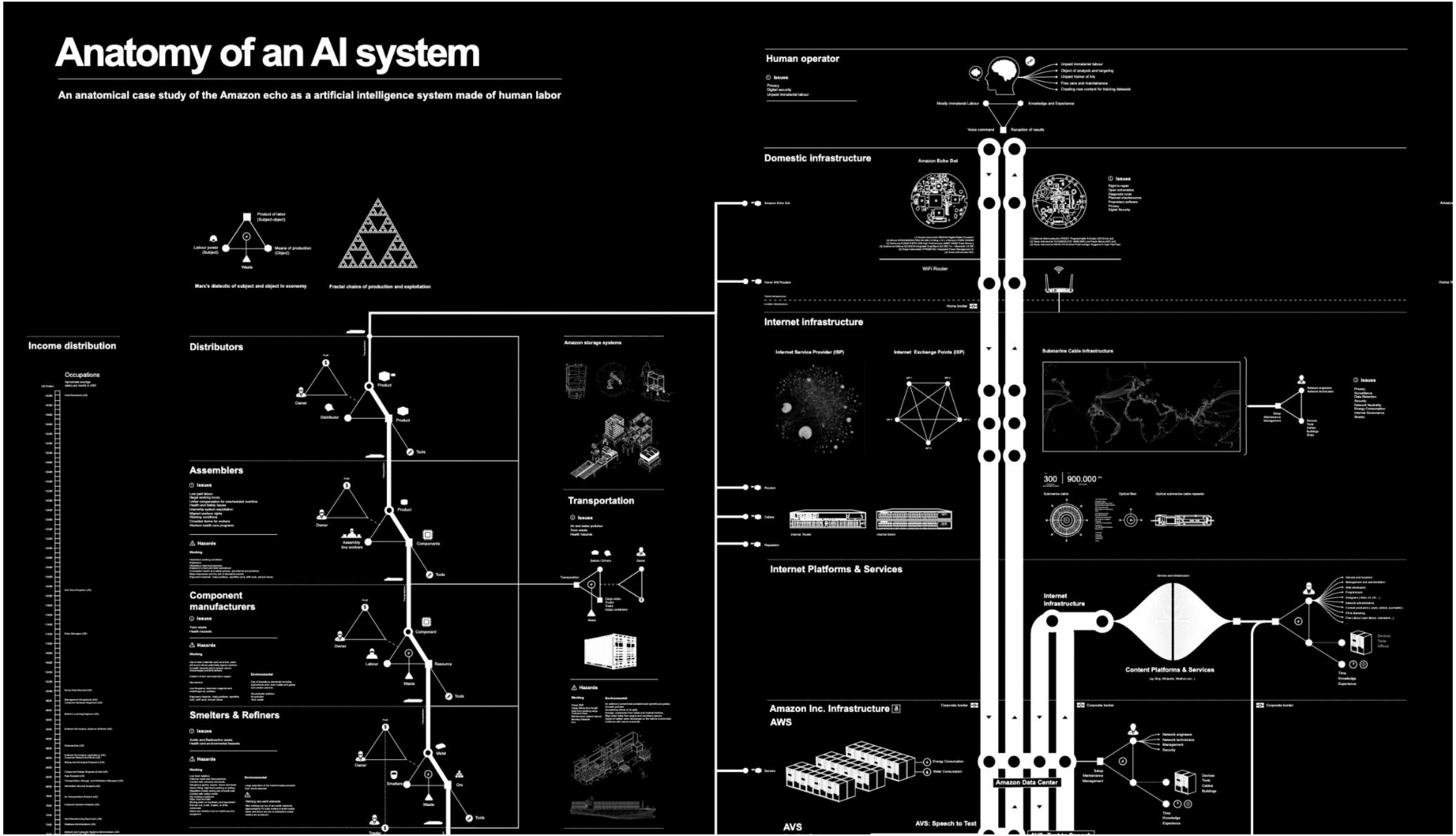
"To me programming is more than an important practical art. It is also a gigantic undertaking in the foundations of knowledge."

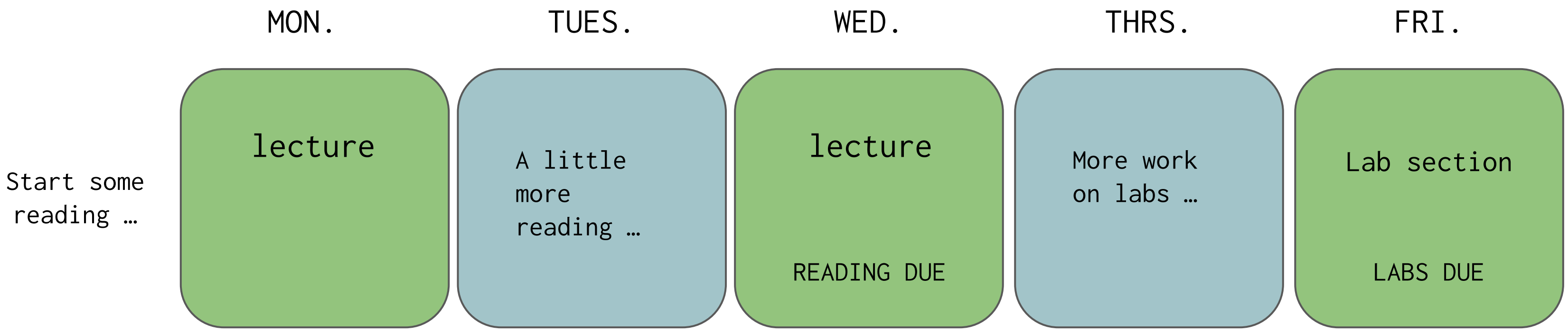
McKenzie Wark, A Hacker Manifesto:

“Hackers create the possibility of new things entering the world.”

Ursula Franklin (1989 lecture):

“As I see it, technology has built the house in which we all live. The house is continually being extended and remodelled. More and more of human life takes place within its walls, so that today there is hardly any human activity that does not occur within this house. All are affected by the design of the house, by the division of its space, by the location of its doors and walls. Compared to people in earlier times, we rarely have a chance to live outside this house. And the house is still changing …”


Anatomy of an AI system
An anatomical case study of the Amazon echo as a artificial intelligence system made of human labor
Human operator
Domestic infrastructure
Amazon Echo Dot
WiFi Router
Internet infrastructure
Internet Service Provider (ISP)
Internet Exchange Points (IXP)
Submarine Cable Infrastructure
300
900.000
Internet Platforms & Services
Internet Infrastructure
Content Platforms & Services
Amazon Inc. Infrastructure
AWS
Amazon Data Center
AVS
AVS: Speech to Text
Marx's dialectic of subject and object in economy
Fractal chains of production and exploitation
Income distribution
Occupations
Distributors
Assemblers
Component manufacturers
Smelters & Refiners
Amazon storage systems
Transportation


MON.
TUES.
WED.
THRS.
FRI.
Start some reading …
lecture
A little more reading …
lecture
READING DUE
More work on labs …
Lab section
LABS DUE

How to do well in this class?

- Come to **class** - usually, without devices

- Be **patient** and **curious** abou the **process** of programming

- ⏰ Give yourself enough **TIME**!

LAB = hackerspace?

2 hackathons in class

1 final project

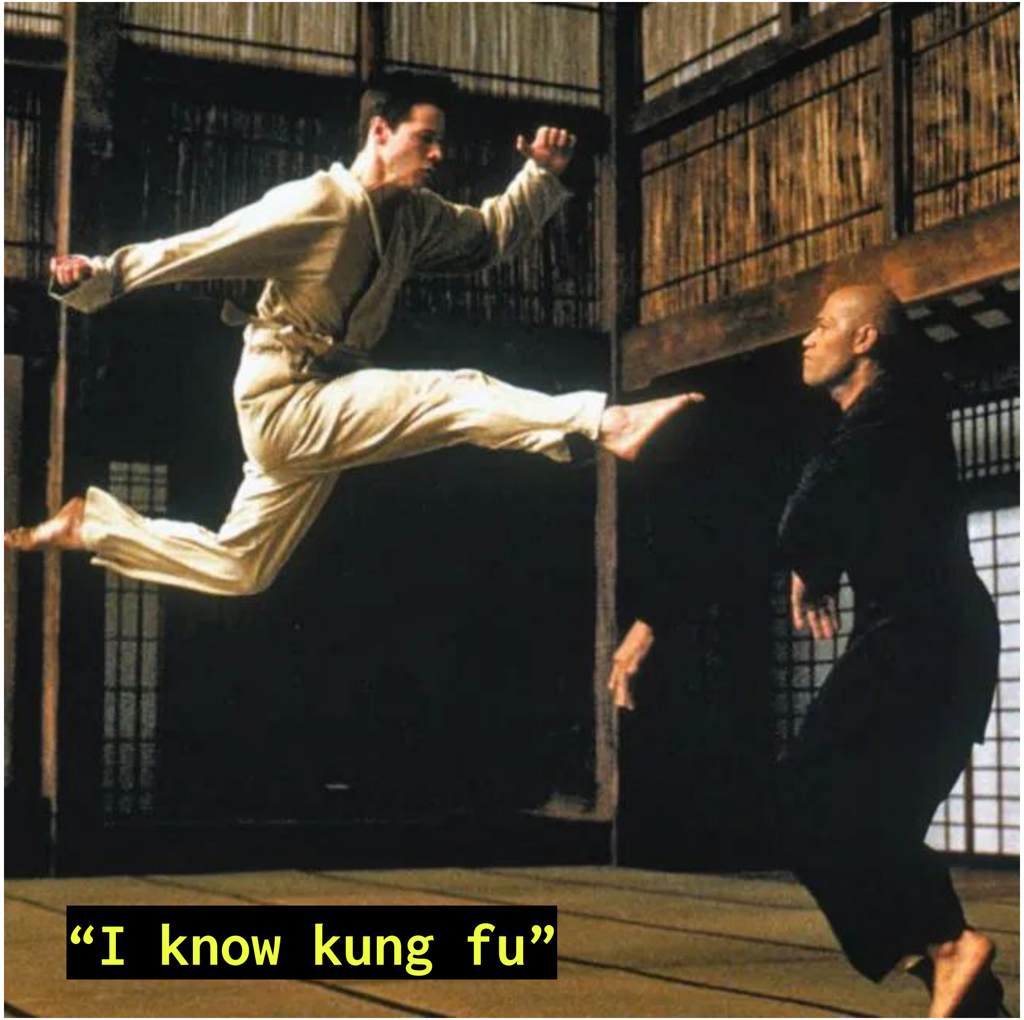
No exams

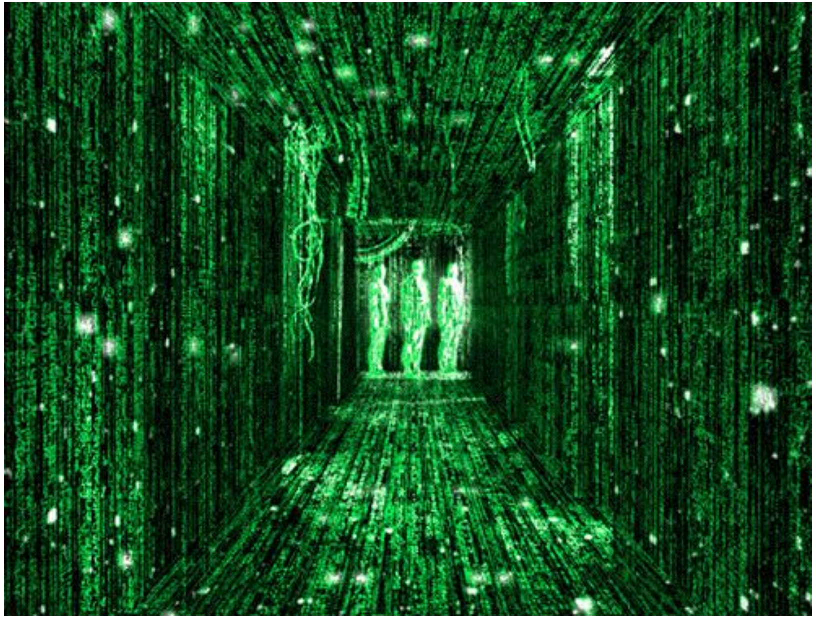
“I was taught … that one ought to figure out a program completely on paper before even going near a computer. I found that I did not program this way … instead of patiently writing out a complete program and assuring myself it was correct, I tended to just spew out code that was hopelessly broken, and gradually beat it into shape. Debugging, I was taught, was a kind of final pass where you caught typos and oversights. The way I worked, it seemed like **programming consisted of debugging**.

If I had only looked over at the other makers, the painters or the architects, I would have realized that there was a name for what I was doing: <u>sketching</u> … **You should figure out programs as you're writing them, just as writers and painters and architects do.**”

-Paul Graham, software engineer + writer


Blu-ray Disc
1080p
Upscaled to
4K


"I know kung fu"



```cpp
#include <iostream>
using namespace std;

int main() {
  cout << "Hello World!" << endl;
}
```

<u>Week 0: Assignment</u>

- **<u>Read syllabus!</u>**
- Instructions for this first set of tasks are in there
- Get your zyBooks! Start Week 1
- Log into Discord - invite soon
- Read class survey
- **<u>SURVEY DUE: Monday Aug. 31st @ 11:59pm</u>**

SEE YOU NEXT LECTURE: <u>Monday Aug. 31st</u>